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Torque Responsive Locking Mechanism for Closed System Transfer Devices

Research Proposal

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Submission Date	20 September 2026

International Academic Research Competition

Research Problem

Anti-neoplastic drugs have widespread use in the cancer industry because they damage or kill cells that rapidly divide. However the properties that make these drugs effective against fighting cancer are also harmful to healthy cells. Many hazardous or toxic drugs used for cancer treatment have cytotoxic, mutagenic, carcinogenic, or reproductive effects, making their safe handling important for healthcare workers, patients, family members, caregives, and others who may come in contact with the drug or contaminated surfaces. The National Institute of Occupational Safety and Health(NIOSH) states that pharmacists, nurses, or any other healthcare workers who handle these hazardous drugs are exposed during daily work. Exposure to these drugs in the long term can lead to reproductive problems, organ damage, and increased risk of some forms of cancer. According to NIOSH, approximately 8 million health care workers in the United States are exposed to these toxic drugs in the workplace. [1]-[3]

During preparation or administration as a droplet, liquid, aerosol, or vapour, drugs can escape and exposure can occur. Because of this protective equipment and biological safety cabinets are important but they are not the only controls that should be enforced . Currently there are several studies that showcase occupational exposure and contamination on surfaces even when these precautions are used. This highlights the important need for an engineering solution that reduces the chances of hazardous drugs escaping during drug handling and administration. Pre-relevant in India because the use of CSTD is not consistently established across the healthcare settings because it can be quite costly. Indian hospitals rely on combinations of biological safety cabinets, PPE, and careful handling procedures, but they do not completely remove the possibility of contamination during drug transfer. The usage of a device like this is especially important as cancer treatment expands across the country and more healthcare workers, patients, and caregivers may come in contact with hazardous drugs. Developing safer and more practical drug transfer systems could be very helpful to strengthen handling practices in any developing country like India. [1], [2]

Closed System Transfer Devices

Closed system transfer devices are one such solution that has been very effective. NIOSH defines a CSTD as a device that transfers drugs but it also mechanically prevents environmental contaminants from entering the system and it also prevents hazardous drug or vapour from escaping it. A CSTD consists of several connected components, including a valve adapter, a syringe adapter, and an administration component. Together, these components create a closed pathway through which the drug can be transferred to the patient. [2]

Connection Security and Accidental Disconnection

A problem for a CSTD to work as intended, however, is containing the vapour or liquid inside the device. Throughout drug preparation and administration, the individual components must also remain securely connected. A CSTD can only remain a closed system when its components stay joined. If a component like a syringe adapter or any other connection separates suddenly, the “closed environment” is lost.

This becomes an important factor, as during normal use syringes are not held completely still. They are constantly re-connected, rotated, and moved while drugs are withdrawn, transferred, or administered. While they are disconnected the syringe’s tip along with toxic material remains exposed. Such movements apply rotational force or torque to the connection. If reverse torque is transferred directly to a connection such as the Luer lock, it begins to loosen the connection, causing accidental leakage. Accidental disconnection, however, is not the only theoretical possibility. FDA MAUDE has reported many cases in which healthcare workers have complained about unexpected separation of CSTD components, including reports that involve drug spills. The reports don't show how frequently accidental disconnection happens and do not prove that a particular commercial device has a design defect. Although they show that unintended separation is actually a failure during clinical use, this is vital because even a CSTD with a very effective filtration system and vapour containment can't perform as a closed system if its components physically separate. [5], [6]

Existing Literature

Most published papers of CSTD studies only focus on the overall performance of the complete CSTD system used commercially. Up till now, Researchers have tested: Hazardous drug contamination, Vapour release, Sterility, Material compatibility, Repeated use,and Drug loss. For example, Wilkinson et al. measured the amount of a

surrogate chemical released during simulated preparation and administration of the drug to compare several commercial CSTDs with a conventional needle and syringe. Each system showed clear differences from one another, and it showcased the importance of testing whether toxic material can cross the system's boundary. However, currently there is less focus on one specific problem: the relationship between the rotational freedom and the resistance to accidental disconnection. Both of these requirements work against each other. If a connector is completely fixed and is against rotation, movement of the syringe can transfer torque directly to the connection. If the torque acts in the direction used to unscrew the connection, it may lead to loosening. On the other hand, simply making a connection rotate freely is not enough because the components must also remain locked together and resist axial disconnection. Therefore, the problem is to allow the connected parts to rotate after the system is engaged while still preventing the syringe from being suddenly unscrewed or pulled apart. [4]

The problem is addressed in existing CSTD designs in various ways. The BD PhaSeal Optima system Uses a locking mechanism to secure the connector to a Luer device. This device also has a spinning collar that allows the syringe to rotate while it is connected. This feature was mainly included because it allows syringe markings to remain visible and reduces twisting of the attached IV lines. This feature shows that controlled rotation can be useful because it shows that the movement of the syringe does not have to necessarily be transferred to the entire device. [7]

The Chemfort Syringe Adaptor Lock uses a different approach. It contains a patented locking mechanism and is designed to fit a Luer-lock syringe intended to prevent accidental disconnection. According to the manufacturer, once it is locked, it can normally not be reopened. This provides a strong connection between the syringe and the adapter. Although, it also means that the syringe-side connection is intentionally difficult to separate once it is locked. EQUASHIELD has a more unified system with its syringe units being part of the closed drug transfer system and not just based on a conventional detachable syringe connection. It also features lower lock adapters to link to standard access points. Equashield owns patents disclosing connector arrangements that may allow a syringe and connector to swivel relative to each other about their common axis. [8], [9]

These examples highlight that there is no single method used to manage rotational forces in CSTD connections. Some systems make the syringe connection difficult or impossible to reopen, while some others introduce a rotating component. For this study, the important point was that rotation itself is not necessarily harmful. From this, we know that the important question is where that rotation occurs and whether it causes relative movement between the male and female Luer threads.

One example relevant to this is Simplivia's WO2017125920A1 Luer Lock Adapter. They have created a lower lock element that behaves differently depending on which direction it is rotated in. When the syringe is attached, ratchet-like teeth restrict rotation of the internal lower element in the tightening direction. Because of this, the external male luer of the syringe can be screwed tightly into the internal female luer. However, when the syringe is rotated in the opposite direction, the internal luer element is able to rotate with the syringe. The pattern specifically describes this common rotation as a way of preventing the male luer from disconnecting from the female luer. [10]

The important idea found in these existing designs is the separation of axial movement from rotational movement. Axial movement refers to the movement along the connector's length. The connection may be separate when two connected components can move too far apart axially. Rotational movement, however, occurs around the central axis of the connector. A component can be prevented from moving either forward or backward while still remaining free to rotate. [11]

Research Question Hypothesis and Original Contribution

Therefore this study investigates whether this problem can be addressed through a torque responsive locking mechanism. First, existing CSTD designs and patents were analysed and studied to understand the different connection and locking systems already in use. Secondly, a new syringe adaptor mechanism was then designed in Creo. The proposed design has a needle hub, spring, locking ribs, and a circumferential groove. During connection, the needle hub moves forward until the locking ribs enter the groove. Once the ribs reach the groove, the hub remains restrained in the axial direction, yet it also is free to rotate within it. The main idea here is that when rotational force is then applied in either the clockwise or anticlockwise direction, the entangled parts rotate together as one unit. Reverse torque is therefore not just directly converted into relative rotation between the syringe and adapter that could unscrew the connection.

Can a torque-responsive locking system allow the connected parts to rotate in both directions after engagement while still preventing reverse twisting from accidentally loosening the syringe adapter connection in a CSTD?

Therefore, It was hypothesised that once the needle hub enters the circumferential groove, rotational force applied to the syringe would cause the syringe and needle hub to co-rotate. This would reduce relative rotation at the Luer interface and therefore reduce the tendency for accidental unscrewing. The proposed mechanism is designed to balance two requirements: rotational freedom during handling and resistance to accidental disconnection.

Design Development

The requirements from the literature review were used as the starting point for developing the proposed system. The design focuses on controlling how the needle hub moves at different stages: while the syringe is being connected and after the connection is complete.

This led to the foundational design in which the type of movement of the needle hub changes depending on its position within the housing. Initially, the hub must remain restricted from rotating so that the syringe can be tightened easily. After the Luer connection has been secured though, the hub must move into a different position where it can rotate freely while still remaining inside the housing.

To achieve these requirements, a new mechanism was developed between the needle hub and the housing. The needle hub contains the female Luer connection and locking ribs on its outer surface. The housing contains internal ribs, a flange, a projection and a circumferential groove. Each of these features controls a different part of the hub's movement and allows the mechanism to change between its two operating positions.

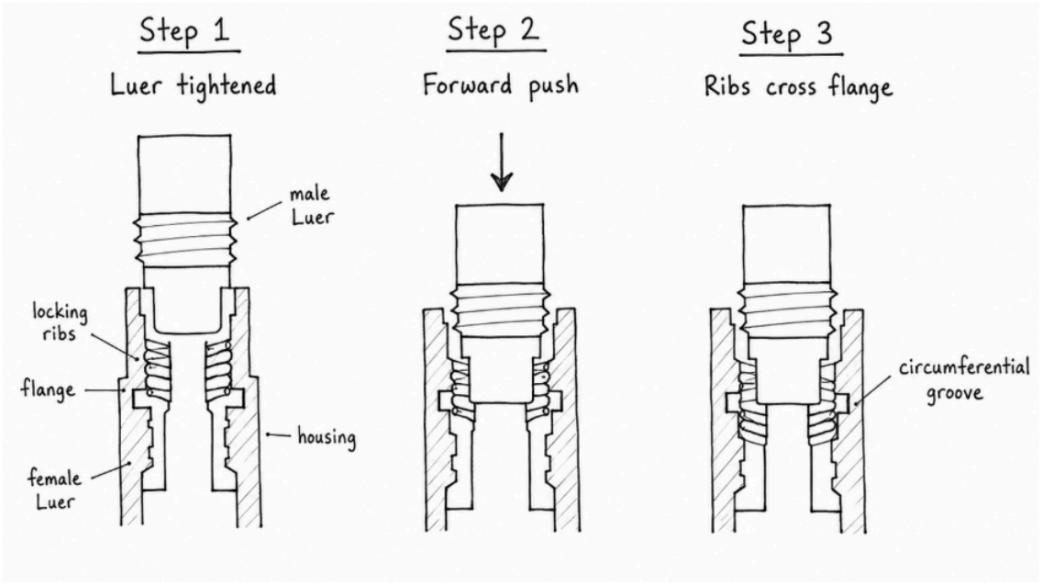


Figure 1 Transition into the Second state

As soon as the locking ribs reach the circumferential groove, the hub will reach its final position. The structure of the housing unit will prevent the hub from moving backwards and returning to its original position. At the same time, the circular groove provides space for the locking ribs to move around the inside of the housing unit. Because of this, the needle hub will be able to rotate while remaining secured in the final position.

Methodology

Once the design of the mechanism had been finalised, the individual components were then modelled in Creo. The CAD model was used to define the needle hub, the housing unit, which included the internal ribs, flange, and circumferential groove that control the movement of the hub.

The components were 3D printed and then assembled into a physical prototype after the design was completed. The design prototype was used to represent only the two-state mechanism that was designed and check whether the parts could be assembled in an intended arrangement.

When connection occurs, the housing unit restricts the rotation of the needle hub so that the syringe can screw into it. After it is tightened, though, the syringe is pushed forward and the hub crosses the flange into the circumferential groove. At this position, the hub is held inside the housing unit but can also rotate clockwise or anticlockwise.

The main purpose of this section is to analyse why this change in movement could reduce accidental disconnection. Since this is a medical device, this analysis is theoretical and does not represent measured prototype performance.

Mechanical and Theoretical Analysis

A screw thread can be represented and approximated by a helix. A screw thread can be thought of as a spiral running around the connector. As the thread will turn, this spiral causes the connected part to move either forwards or backwards. The distance it moves along the connector to complete one turn is called the lead or L. For a single-start thread, this is the same as the thread pitch(distance between two neighboring thread ridges). So if the thread lead is represented by L, one full rotation of 2π radians produces an axial displacement of L. Therefore, the ideal relationship between the relative rotation and axial movement is: [12]

Friction can be a constraint, along with manufacturing clearance. Deformation and resistance between the hub and housing may prevent perfect co-rotation. The following model establishes the mechanical condition that the design tries to follow.

A real hub is most likely not going to follow the syringe perfectly. To account for this, a rotation transfer coefficient should be introduced:

$$k = \theta\square / \theta\square$$

$$\Delta\theta = \theta\square(1 - k) \qquad x = (p\theta\square / 2\pi)(1 - k)$$

$$T\square < T_u$$

This model produces a theoretical prediction This is theoretical data rather than the experimental result because the actual value of k has not been measured yet for this prototype.

Manufacturing Considerations

The current design or prototype was produced by a 3D printing machine, which is suitable for demonstrating the proposed design, but it does introduce dimensional irregularities. Manufacturing accuracy can depend on factors such as build direction, layer thickness, and parameters, which are relevant for small and tonal features such as the ribs and the grooves. Experimental NIST work on polymer fused filament fabrication has shown that printing conditions can affect both the design and the mechanical properties of the prototype. [13]

This is important because the mechanism needs the right amount of space between the hub, flange, and the circumferential groove. If there is very little space, it increases friction and makes the hub harder to rotate, while if there is too much space, it can cause too much movement, which is unwanted.

If the parts were injection moulded, they would have different dimensional and mechanical behavioural properties from a 3D printed prototype. Polymer additive manufacturing studies have found that interlayer bonding can cause mechanical behaviour to differ from injection moulded material. It would not be valid to assume ,however, that injection moulding would automatically increase the disconnection force that would need to be verified using production quality samples. [14]

Expected Outcome

This theoretical analysis predicts that the mechanism will be most effective when the hub is securely held against axial separation but can rotate with less resistance than is required to unscrew the tightened luer connection. This prediction has provided the basis for the results and future experimental tests discussed in the following section.

The model developed in section 4 showed that the amount of thread movement depends on how closely the needle hub follows the rotation pattern of the syringe.

When the hub is fixed, the syringe's rotation occurs relative to the female Luer. This will create the greatest possibility for the male luer to move backwards through the threads when the syringe is turned in the loosening direction.

As more of the syringe's rotation is passed on to the needle hub, the difference in rotation between them becomes smaller. For example, the theoretical model above predicts that if the hub follows 50% of the syringe's rotation, 50% of the original relative rotation will remain. If it follows 75%, only 25% remains, while 90% rotation transfer leaves only 10%.

Therefore, the main result is that the mechanism changes where the rotation occurs. After engagement occurs, the movement is the rotation between the needle hub and the housing unit rather than the rotation between the male and female Luer threads.

For an ideal case where the hub rotates through exactly the same angle as the syringe, there would be no relative angular movement between the two Luer components. Therefore, from the theoretical model, we know that there will be no thread-driven backward movement in the ideal condition.

However, since the prototype is not made with manufacturer-grade materials, the actual rotation transfer coefficient of the prototype has not been measured. Friction, manufacturing tolerances, and deformation may prevent the hub from following the syringe perfectly, and the calculated values therefore represent calculated behaviour rather than actual device performance.

Limitations and Future Validation

The main limitation of the study is that the proposed design has only and mainly been evaluated through CAD development of physical proof of concept and theoretical mechanical analysis. The prototype has not undergone control testing for leakage, sterility, or hazardous drug containment, so no conclusions can be made about its clinical performance as a complete CSTD.

The main limitation is that the current prototype represents a very early-stage model, as it's not production quality yet. The theoretical analysis assumes idealised behaviour between the components, while the real mechanism will also be affected by factors like friction, material deformation, and wear. Therefore, while the model predicts how increasing co-rotation could lead to disconnection accidentally, the magnitude of this effect in a physical working prototype has to be determined accurately.

Similarly, It was identified how much pulling force would be transferred through the locking features, but the maximum failure load has not been established yet. Accurate material properties and experimental loading would be required before we could develop a reliable safety factor.

Validation from a clinical setting would be the next stage of development. The final CSTD would require production, great quality materials, and a manufacturing unit, followed by testing for fluid leakage, sterility, and hazardous drug containment under relevant medical device requirements. [15]-[17]

The present study does not establish clinical readiness. It instead develops and analyses the mechanical principles that would justify progressing it to the later validation stages. The theoretical predictions presented show measurable targets for future testing, which allows the proposed mechanism to be evaluated against its intended function.

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